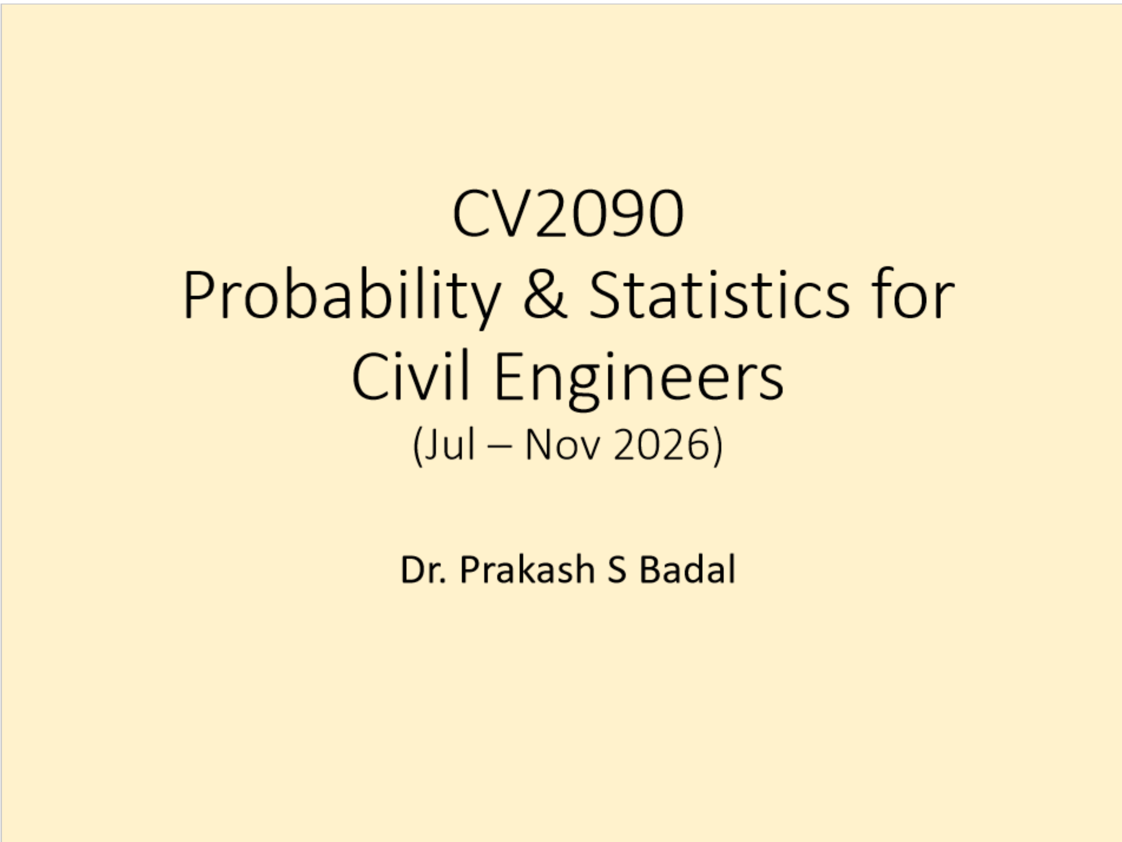
inClass-L9

Wednesday, August 12, 2026 11:22 AM

inClass- L9



Poisson rv

- Given λ: average number of successes in a time window
- X is the rv for the number of successes
- The PMF is

$$p_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0,1,2, \ldots \,\&\, \lambda > 0$$

- Is this a legitimate PMF?
- Show that $\sum_k p_X(k) = 1$

L9
Aug 13, 2026

Traffic Flow

- **Q:** Highway capacity?
  - Maximum **vehicles one lane of a highway can carry in an hour** without congestion?

https://ladingcargo.com/blog/planning-your-drive-avoiding-heaviest-traffic-highways-in-2025/

speed = 100 kmph
length of veh ≈ 5 m
headway ≈ 2-s headway
front-to-front dist = 5 m + 55 m = 60 m
vehicles /km = 1000/60 ≈ 16.7 veh/km
density ↗
flow = speed x density ≈ 1670 veh/h/lane

5000 veh/h/lane
20,000 veh/h/lane

very close to real life data

1800 - 2200 veh/h/lane

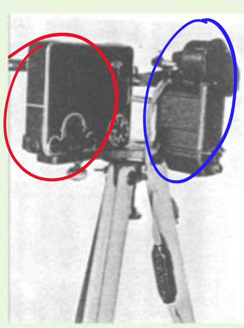
4

Traffic Flow Theory

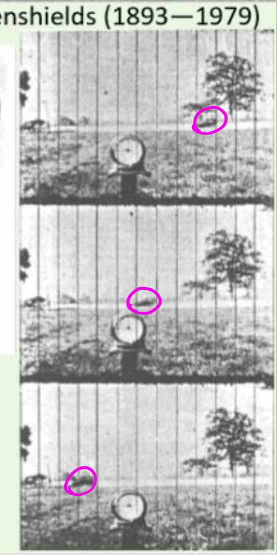
- **Q:** How can we increase capacity?

→ Increase speed X
→ Reduce speed X
→ Both ?
Q: Is every vehicle @ at exact 60 m headway?

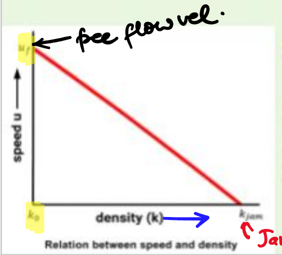
Bruce Greenshields (1893—1979)



Camera with clockwork motor



on-field data → parameters.



free flow vel.

Jam density

Greenshields' model

$$q = kv$$

k: density, veh/km
v: velocity, km/h
q: flow, veh/h

$$v = v_f - \left(\frac{v_f}{k_{jam}}\right)k$$

Traffic Flow Theory

→ What is the fraction of vehicles that are within 50m headway?

CDF

Cumulative Distribution Function

6

Messages from the Last Class

- **Message 1:** Discrete RVs need a "Probability Mass Function"

→ Bernoulli rv/trial

$$p_X(x) = \begin{cases} p, & x = 1 \\ 1-p, & x = 0 \end{cases}$$

{ single param model → p

→ Binomial rv

X is number of successes in n Bernoulli trials

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0, 1, 2, \dots, n$$

two param n, p

→ Geometric rv

X is number of trials required for first success

$$p_X(x) = (1-p)^{x-1} p$$

→ Poisson rv

X is number of success in a given time window

$$p_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

7

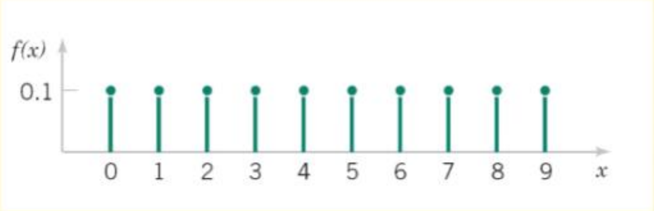
Poisson rv

- For a Poisson distribution, when are the two consecutive probability masses equal?

8

Discrete Uniform rv

$$p_X(x) = \frac{1}{n} \qquad x = 1, 2, \cdots, n$$



9

Bridge Bearings

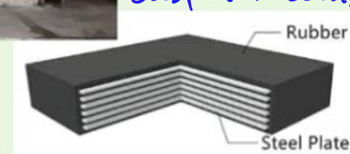
State of TN, has 5,000 major highway bridges.

Average life of Bearing ≈ 20 years.

How many bearings would require replacement in the next five years?

↳ CDF

Elastomeric Bearing



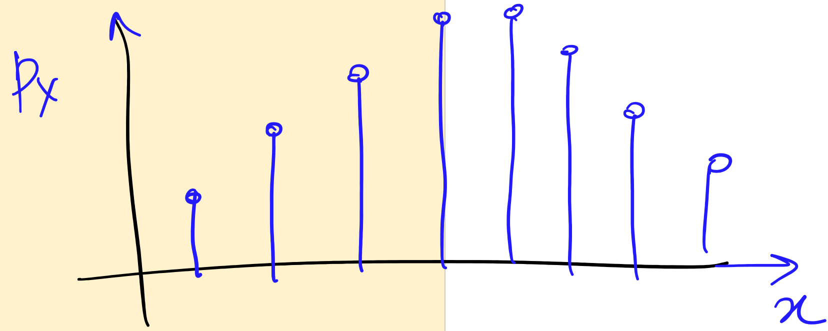
CDF

Cumulative Distribution Function (CDF)

Definition

$$F_X(x) = Pr(X \leq x)$$

$$F_X(x) = \sum_{x_i \leq x} p_X(x_i)$$



Properties

$$F_X(-\infty) = 0$$

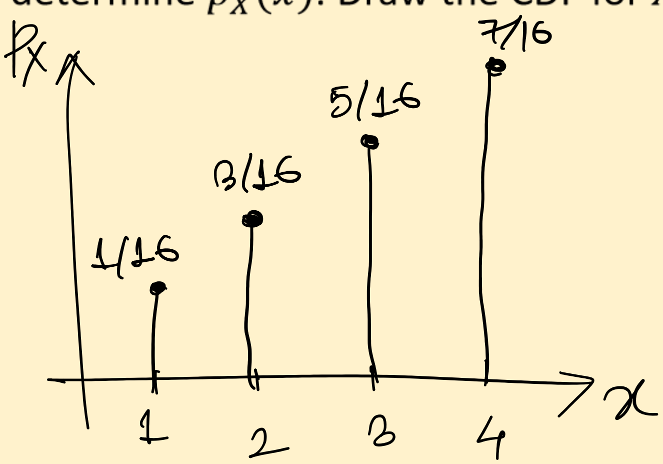
$$F_X(+\infty) = 1$$

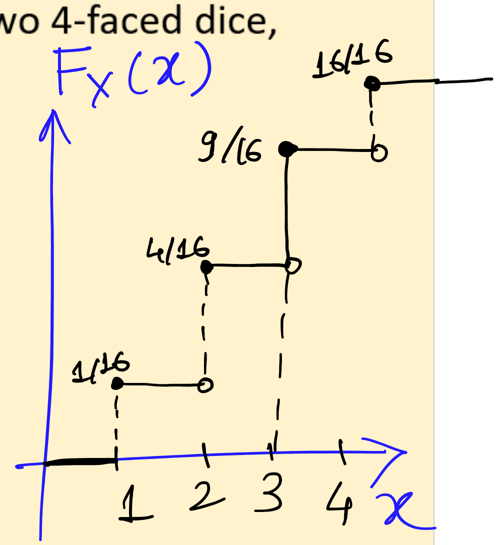
if $x < y$, $\quad F_X(x) \leq F_X(y)$

12

Class problem on CDF

X is the maximum value from rolling two 4-faced dice, determine $p_X(x)$. Draw the CDF for X.





13